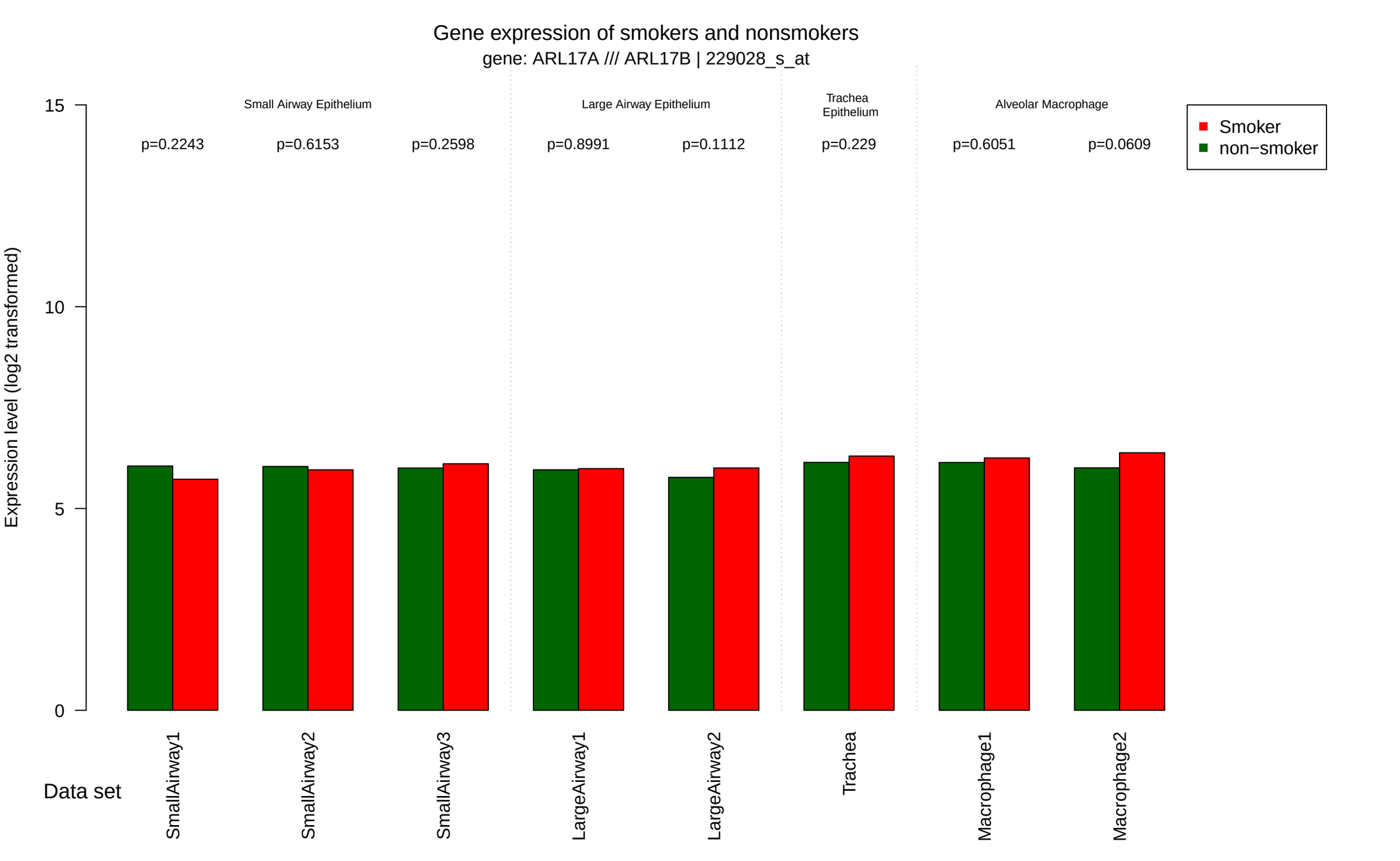
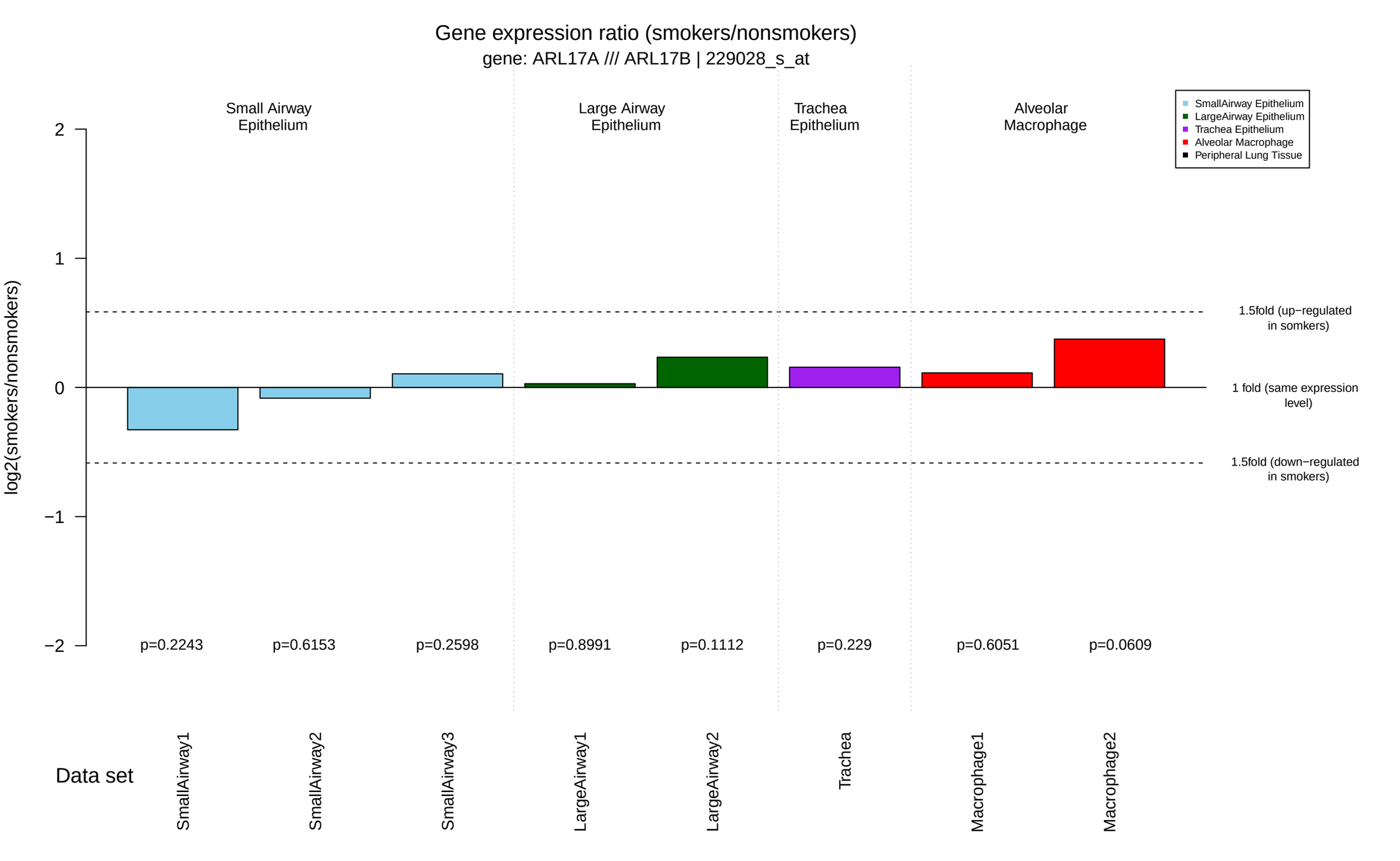
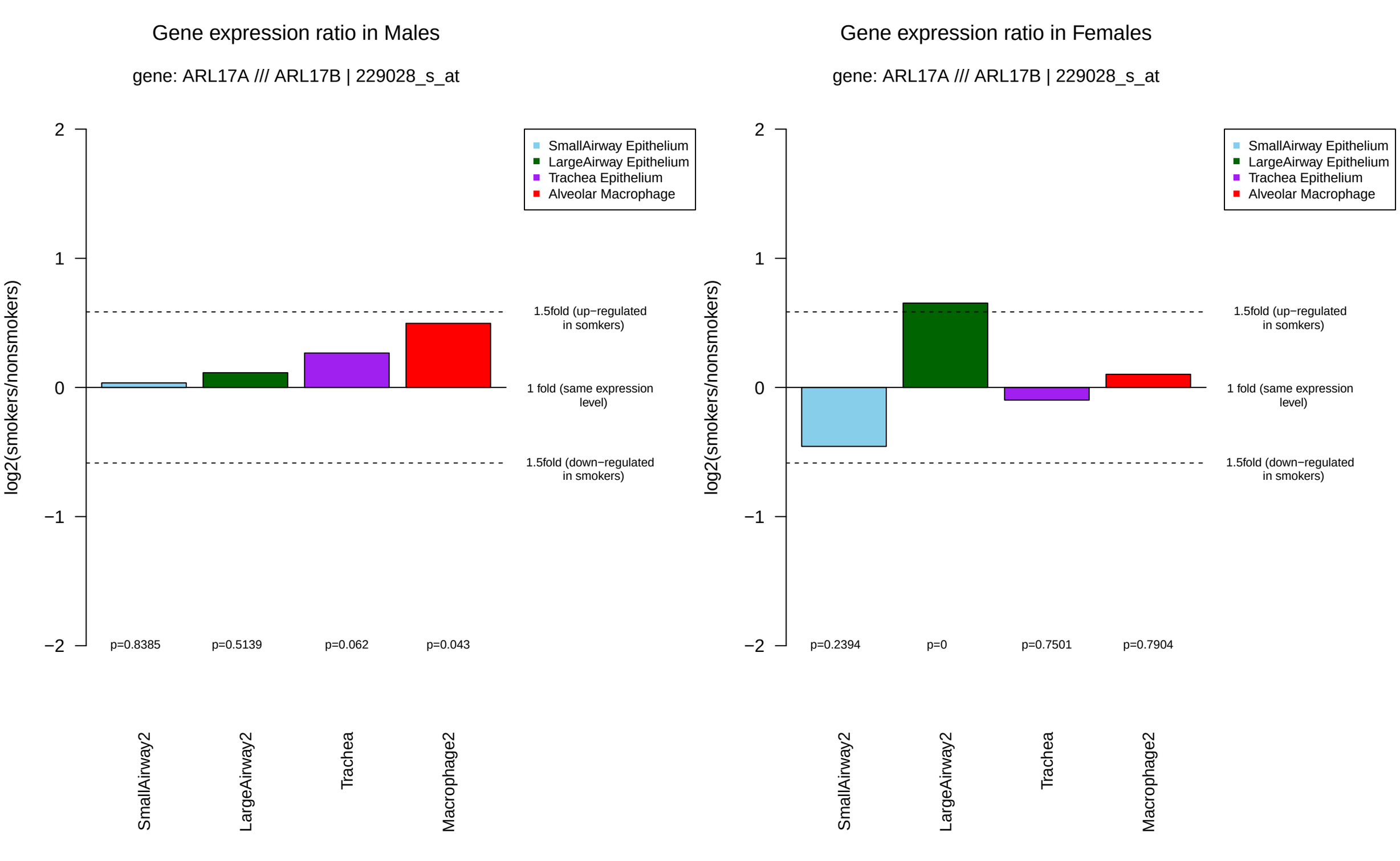
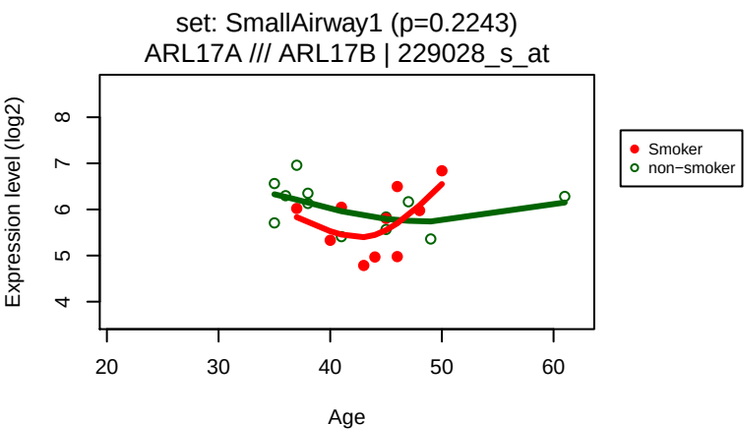
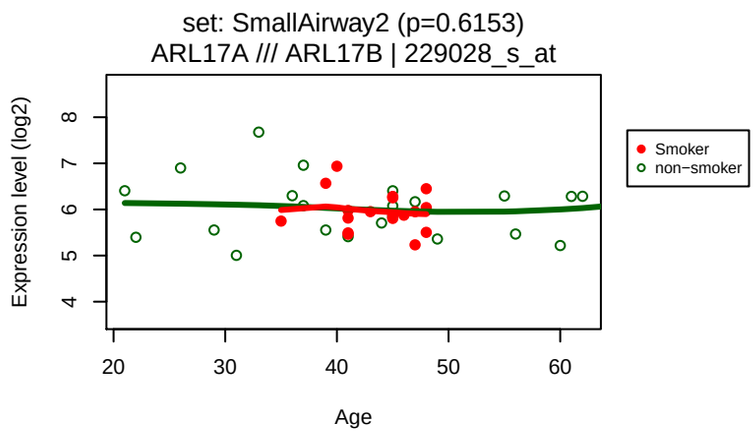
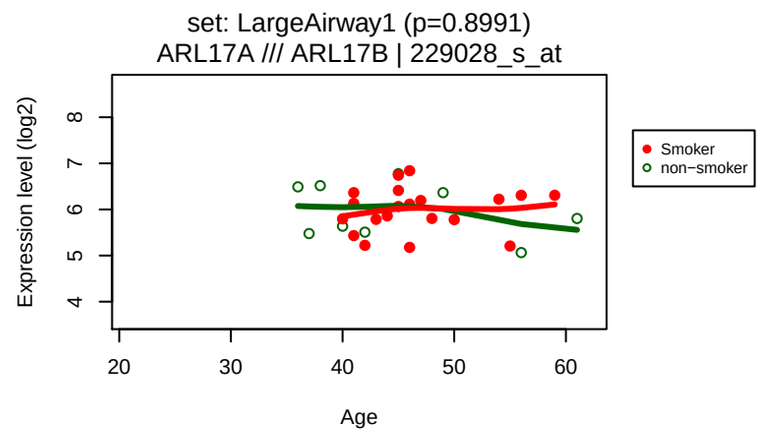

Gene expression of smokers and nonsmokers
gene: ARL17A /// ARL17B | 229028_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2243
p=0.6153
p=0.2598
p=0.8991
p=0.1112
p=0.229
p=0.6051
p=0.0609
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

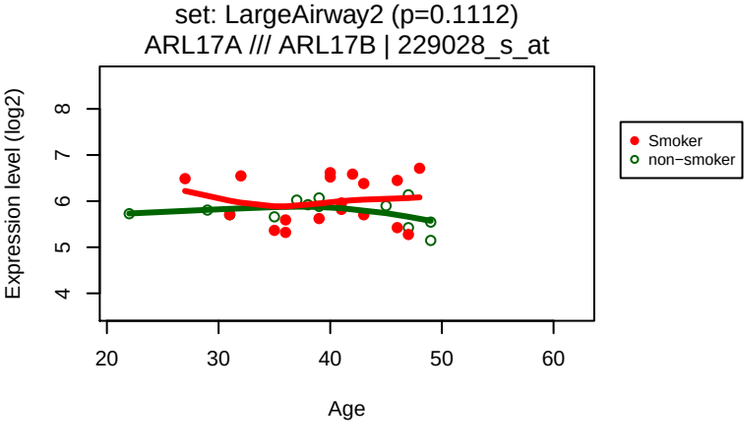

Gene expression ratio (smokers/nonsmokers)
gene: ARL17A /// ARL17B | 229028_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2243
p=0.6153
p=0.2598
p=0.8991
p=0.1112
p=0.229
p=0.6051
p=0.0609
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

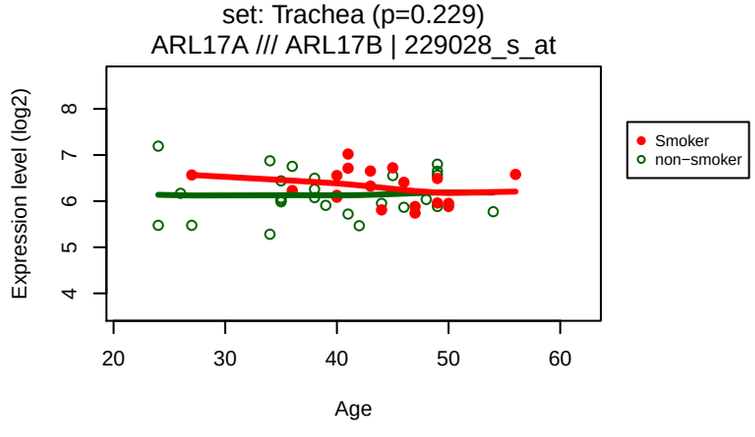

Gene expression ratio in Males
gene: ARL17A /// ARL17B | 229028_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8385
p=0.5139
p=0.062
p=0.043
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL17A /// ARL17B | 229028_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2394
p=0
p=0.7501
p=0.7904
SmallAirway2
LargeAirway2
Trachea
Macrophage2

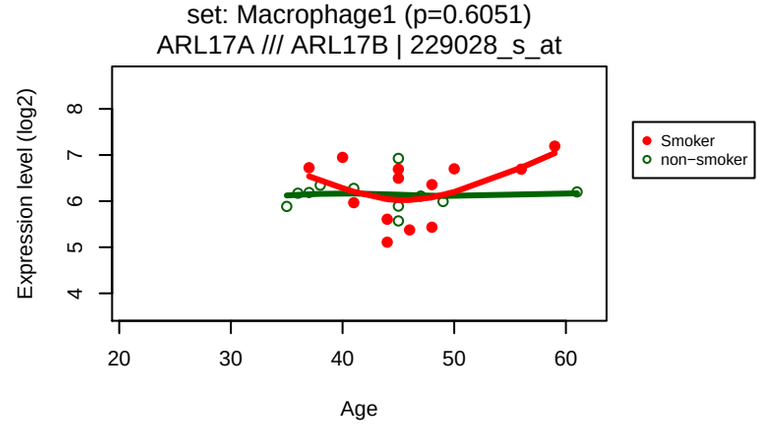

set: SmallAirway1 (p=0.2243)
ARL17A /// ARL17B | 229028_s_at
Expression level (log2)
Age
Smoker
non−smoker

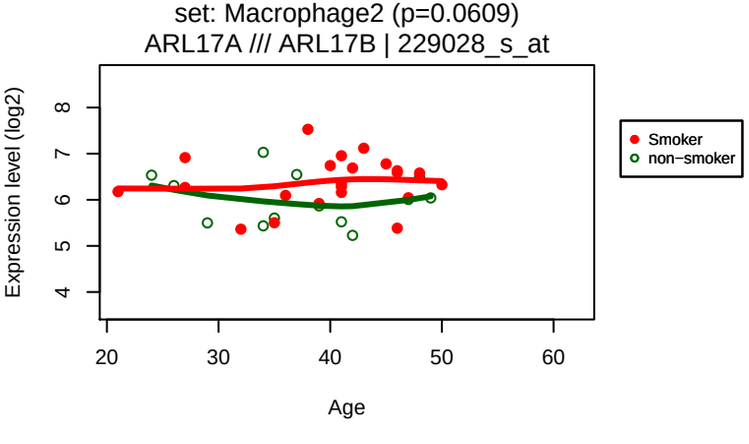

set: SmallAirway2 (p=0.6153)
ARL17A /// ARL17B | 229028_s_at
Expression level (log2)
Age
Smoker
non−smoker

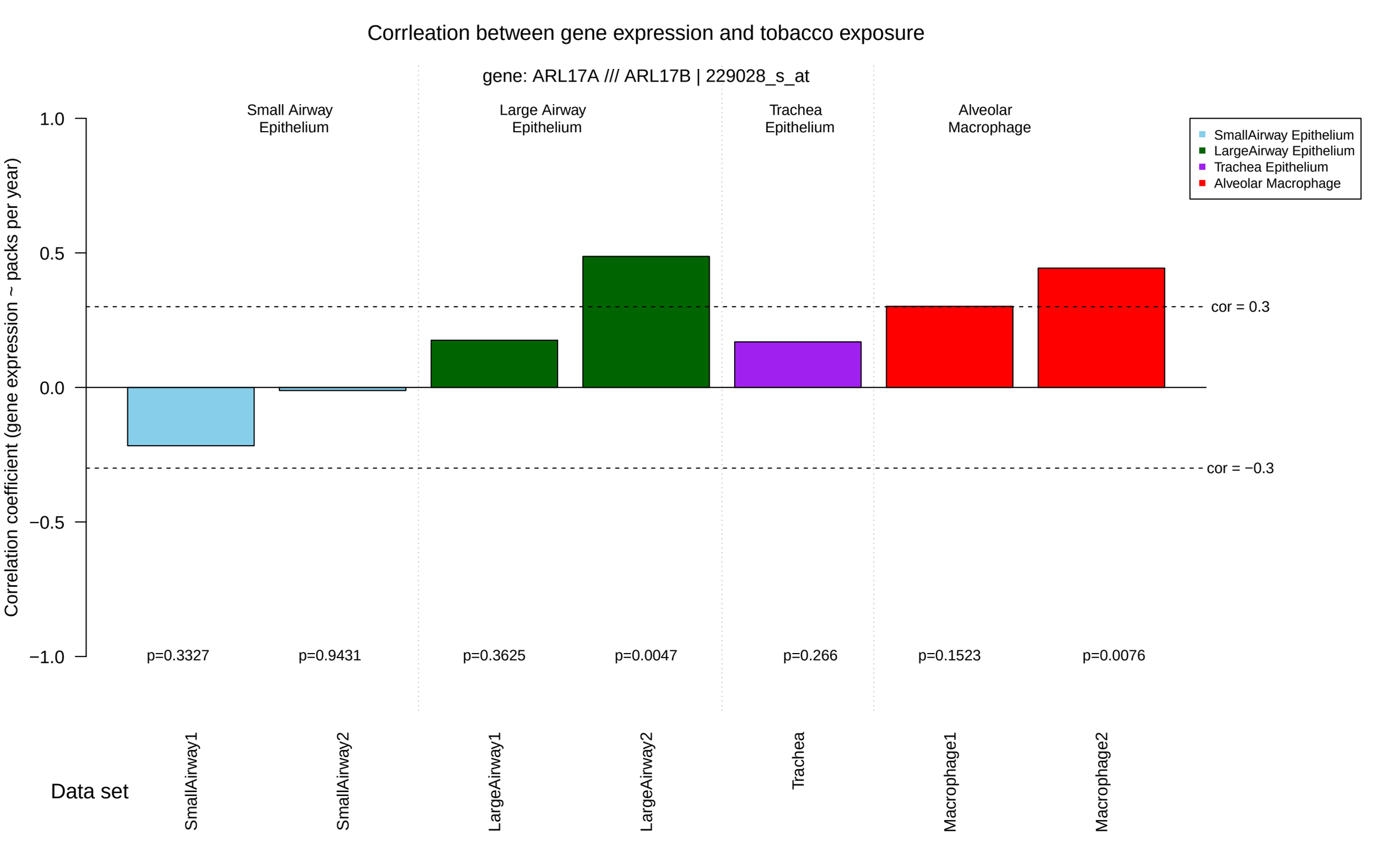

set: LargeAirway1 (p=0.8991)
ARL17A /// ARL17B | 229028_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1112)
ARL17A /// ARL17B | 229028_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.229)
ARL17A /// ARL17B | 229028_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6051)
ARL17A /// ARL17B | 229028_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0609)
ARL17A /// ARL17B | 229028_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL17A /// ARL17B | 229028_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3327
p=0.9431
p=0.3625
p=0.0047
p=0.266
p=0.1523
p=0.0076
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2